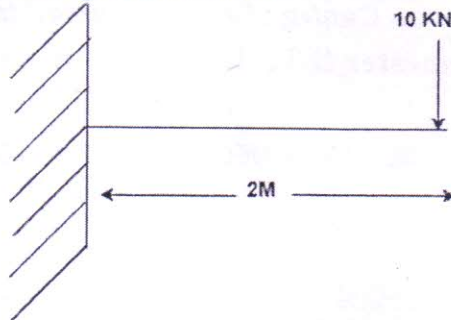
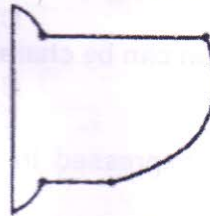


10. Shear force diagram of the following beam will be



- A) rectangle
B) cubic
C) parabolic
D) trapezoidal
11. A simply supported beam has equal over-hanging lengths and carries equal concentrated loads P at ends. Bending moment over the length between the supports
- A) is zero
B) is a non-zero constant
C) Varies uniformly from one support to the other
D) Is maximum at mid-span
12. Moving trains are an example of ____ load.
- A) point
B) kinetic
C) rolling
D) sliding
13. A beam of uniform strength has
- A) same cross-section throughout the beam
B) same bending stress at every section
C) same bending moment at every section
D) same shear stress at every section
14. The layer of zero stress due to bending in beams is called
- A) neutral layer
B) Shear layer
C) central layer
D) natural layer
15. The shear stress distribution over a beam cross section is shown in the figure. The beam is of



- A) Equal flange I-Section
B) Unequal flange I-Section
C) circular section
D) T-section
16. Centroid of a section is
- A) About which $\int y^2 dA = 0$
B) About which $\int y dA = 0$
C) About which $\int xy dA = 0$
D) About which $\int x^2 y^2 dA = 0$
17. A simply supported beam with width 'b' and depth 'd' carries a central load W and undergoes deflection δ at the centre. If the width and depth are interchanged, the deflection at the centre of the beam would attain the value
- A) $d(\delta/b)$
B) $(d/b)^2 \delta$
C) $(d/b)^3 \delta$
D) $(d/b)^5 \delta$
18. The maximum deflection in a cantilever beam of span "l" m and loading at free end "W" kN is
- A) $Wl^3/2EI$
B) $Wl^3/3EI$
C) $Wl^3/4EI$
D) $Wl^2/2EI$
19. The outside diameter of a hollow shaft is twice its inside diameter. The ratio of its torque carrying capacity to that of a solid shaft of the same material and the same outside diameter is
- A) 15/16
B) 3/4
C) 1/2
D) 1/16

20. What is the polar modulus for solid shaft?

- A) $A(\pi/16) D^4$
C) $(\pi/16) D^3$

- B) $(\pi/12) D^4$
D) $(\pi/16) D^2$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Show that the extension produced due to self-weight of a bar of uniform circular section fixed at one end and suspended vertically is equal to half the extension produced by a load equal to self-weight applied at the free end.	06	L*2	01	01
	b) Compare ductility and malleability, highlighting how each property affects material performance under stress.	02	L2	01	01
	c) A bar of 30mm diameter is tested in tension. It is observed that when a load of 60 kN is applied, the extension measured over a gauge length of 200mm is 0.1mm and contraction in diameter is 0.004mm. Find Poisson's ratio and elastic constants E, G, K.	08	L3	02	02
2.	a) A tapering rod has diameter 'd ₁ ' at one end and it tapers uniformly to a diameter 'd ₂ ' at the other end for a length of 'L'. If modulus of elasticity of the material is 'E', Determine its change in length when subjected to an axial force 'P'.	08	L2	01	01
	b) A 500 mm long bar has rectangular cross-section 20mm × 40mm. This bar is subjected to (i) 40 kN tensile force on 20 mm × 40 mm faces, (ii) 200 kN compressive force on 20 mm × 500 mm faces, and (iii) 300 kN tensile force on 40 mm × 500 mm faces. Find the change in volume if $E = 2 \times 10^5 \text{ N/mm}^2$ and $\mu = 0.3$	08	L3	02	02
3.	a) For the laboratory tested specimen the following data were obtained. Diameter of specimen=30mm, Length of specimen = 200mm, Extension under the load of 150kN = 0.21mm, Load at yield point = 250kN, Maximum load = 380kN, Total extension = 60mm, Neck diameter at failure = 22.5mm. Determine (i)Young's Modulus (ii)Yield stress (iii)Ultimate stress (iv)Percentage Elongation (v)Percentage reduction in area.	08	L3	01	02
	b) Explain the concepts of Bulk Modulus and Volumetric Strain and illustrate their relationship.	04	L2	02	01
	c) Explain the behavior of a rubber balloon when inflated by applying the concept of volumetric strain.	04	L3	02	02
Unit – II					
4.	a) Differentiate between a sagging bending moment and a hogging bending moment and explain their effects on structural members.	03	L2	03	01
	b) Derive the relationship between load, shear force and bending moment.	05	L2	03	01
	c) A symmetric I-section has flanges of size 180 mm × 20 mm and its overall depth is 400mm. Thickness of the web is 20 mm. Find the moment of resistance of the section, if permissible stress is 90 N/mm ² , what is the maximum uniformly distributed load the beam can carry over a simply supported span of 6m?	08	L3	04	02

5. a) A simply supported beam AB of 6m span is loaded as shown in fig. 5(a). Draw the shear force diagram with forces as shown.

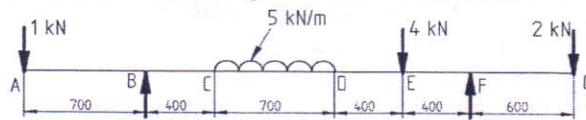


fig. 5(a)

08 L3 03 02

- b) Show that for a rectangular cross section, shear stress distribution varies parabolically across depth. Further show that maximum shear stress is 1.5 times average stress.

08 L2 04 01

6. a) A simply supported beam AB of 6m span is loaded as shown in fig. 6(a). Draw the bending moment diagram when a reaction force at A is 25kN and reaction at B is 50kN. Also indicate the point of contra flexure if any.

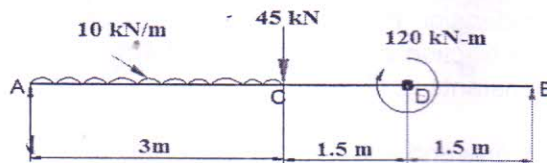


fig. 6(a)

08 L3 03 02

- b) A beam of an "I" section 200mm x 300mm has web thickness 10mm and flange thickness 10 mm as shown in fig. 6(b). It carries a shearing force of 10kN at a section. Sketch the shear stress distribution across the section.

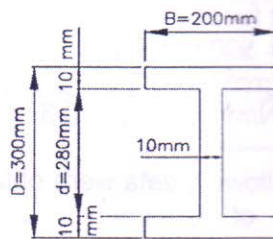


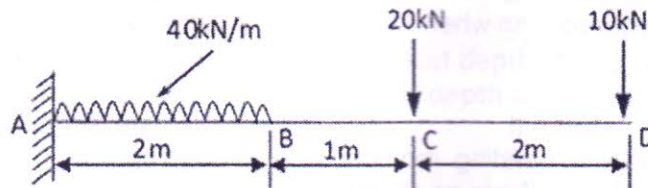
fig. 6(b).

08 L3 04 02

Unit – III

7. a) Derive an expression for deflection for a simply supported beam subjected to UDL.
- b) Find the slope and deflection at the free end of the cantilever shown in Figure. Assume $E=200\text{GPa}$, $I=40 \times 10^6 \text{ mm}^4$

08 L2 05 01



08 L3 05 02

8. a) Derive the relationship between angle of twist and shear stress in a solid circular shaft. Also mention assumptions made
- b) A solid shaft which will transmit 440kW at 280rpm with the angle of twist should not exceed one degree per meter length, and the maximum torsional shear stress is to be limited to 40MPa. Assume $G= 84\text{GPa}$. Determine the diameter of the shaft and analyze considering the strength and stiffness criterion.

08 L3 05 01

08 L4 05 02

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
